Hardware Failure

When a router, switch, NIC, or other hardware goes down, your job as a network technician includes identifying the location of the hardware failure. Even on smaller networks, it can be a challenge to determine exactly which device is causing problems. Though you could manually check each device on your network for errors, you might be able to shorten your list with a little detective work first. Here's how:

1. Use tracert or traceroute (depending on your OS) to track down malfunctioning routers and other devices on larger networks. Because ICMP messages are considered low priority, be sure to run the command multiple times and compare the results before drawing any conclusions.
2. Keep in mind that routers are designed to route traffic to other destinations. You might get more accurate tracert or traceroute feedback on a questionable router if you target a node on the other side of that router rather than aiming for the router itself.
3. As you home in on the troublesome device, use ping to test for network connectivity.

Self-Check

9. You need to determine which device on your network is sending excessive messages to your Ubuntu Server. Which utility will give you this information?
   - **a.** traceroute
   - **b.** tcpdump
   - **c.** netstat
   - **d.** arp
10. What protocol must be allowed through a firewall for tracert to work correctly?
   - **a.** SSH
   - **b.** NDP
   - **c.** ICMP
   - **d.** TLS

☐ Check your answers at the end of this chapter.

Summary

Section 4-1: TCP/IP Core Protocols

- Many TCP/IP protocols function by adding information at the beginning of messages in fields called a header. In a process called encapsulation, a transport layer protocol adds a header to the payload. The network layer adds its own header to the passed-down segment or datagram. The data link layer encapsulates this data with its own header and trailer, creating a frame. The physical layer places the actual transmission on the network. The receiving host decapsulates the message at each layer in reverse order and then presents the payload to the receiving application.
- Each networking device can be described partially by the innermost OSI layer header it reads and processes.
- Before TCP transmits data, it ensures that a connection or session is established. TCP senders and receivers also generate a checksum for each message. If the two checksums fail to match, the destination host asks the source to retransmit the data.
- TCP uses the sequence number and acknowledgment number fields to orchestrate a three-way handshake that establishes a session.
- UDP (User Datagram Protocol) is an unreliable, connectionless protocol, which means that UDP does not guarantee delivery of data, and no connection is established by UDP before data is transmitted.
- IP (Internet Protocol) belongs to the network layer of the OSI model and specifies where data should be delivered, identifying the data's source and destination IP addresses. IP is the protocol that enables TCP/IP to internetwork—that is, to traverse more than one LAN segment and more than one type of network through a router.

- Whereas IP helps direct data to its correct destination, ICMP (Internet Control Message Protocol) is a core network layer protocol that reports on the success or failure of data delivery and other network conditions. ICMP announces these transmission failures to the sender, but it does not correct errors it detects—those functions are left to higher-layer protocols, such as TCP.
- ARP (Address Resolution Protocol) works in conjunction with IPv4 to discover the MAC address of a node on the local network and to maintain a database that maps local IP addresses to MAC addresses. ARP touches information (IP addresses and MAC addresses) at both layer 2 and layer 3. However, it operates only within its local network bound by routers.
- The most important data link layer standard, Ethernet, is adaptable, capable of running on a variety of network media, and offers excellent throughput at a reasonable cost. Unlike higher-layer protocols, Ethernet adds both a header and a trailer to the payload it inherits from the layer above it, creating a frame around the payload.

Section 4-2: Encryption Protocols

- In terms of security, data exists generally in three states: at rest, in use, and in transit. Encryption is the last layer of defense against data theft. Encryption protocols use a mathematical code, called a cipher, to scramble data into a format that can be read only by reversing the cipher—that is, by deciphering, or decrypting, the data.
- To protect data at rest, in use, and in transit, encryption methods are primarily evaluated by three benchmarks: confidentiality, integrity, and availability. These three principles form the standard security model called the CIA (confidentiality, integrity, and availability) triad.
- With private key encryption, data is encrypted using a single key that only the sender and the receiver know. A potential problem with private key encryption is that the sender must somehow share the key with the recipient without it being intercepted.
- With public key encryption, data is encrypted with a private key known only to the user, and it's decrypted with a mathematically related public key that can be made available through a third-party source, such as a public key server. This ensures data integrity, as the sender's public key will only work if the data has not been tampered with. Alternatively, data can be encrypted with the public key, and then it can only be decrypted with the matching private key. This ensures data confidentiality, as only the intended recipient (the owner of the keys) can decrypt the data.
- SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are both methods of encrypting TCP/IP transmissions en route between the client and server using public key encryption technology. SSL operates in the application layer. TLS operates in the transport layer and uses slightly different encryption algorithms than SSL, but otherwise is essentially the updated version of SSL.
- HTTP uses TCP port 80, whereas HTTPS (HTTP Secure) uses SSL/TLS encryption and TCP port 443 to protect data in transit. Other protocols you've studied offer SSL/TLS encrypted alternatives, including SMTPS, IMAPS, POP3S, and LDAPS.
- Each time a client and server create an SSL/TLS connection, they establish a unique session defined by an agreement on a specific set of encryption techniques. The session allows the client and server to continue to exchange data securely as long as the conversation is kept active.
- IPsec (Internet Protocol Security) is an encryption protocol suite that defines a set of rules for encryption, authentication, and key management for TCP/IP transmissions. IPsec works at the network layer of the OSI model—it adds security information to the headers of IP packets and encrypts the data payload.

Section 4-3: Remote Access Protocols

- FTPS (FTP Security or FTP Secure) is an added layer of protection for FTP using SSL/TLS that can encrypt both the control and data channels. SFTP (Secure FTP) is a file-transfer version of SSH that includes encryption and authentication, and it's sometimes inaccurately called FTP over SSH or SSH FTP. Note that SFTP is an extension of the SSH protocol, not of FTP. TFTP (Trivial FTP) is a simple version of FTP that includes no authentication or security for transferring files and uses UDP at the transport layer (unlike FTP, which relies on TCP at the transport layer).
- Terminal emulation, also called remote virtual computing, allows a remote client to take over and command a host computer. Command-line software that can provide terminal emulation includes Telnet and SSH. Some GUI-based software examples are Remote Desktop for Windows and TeamViewer. A host may allow clients a variety of privileges, from merely viewing the screen to running programs and modifying data files on the host's hard disk.
- SSH allows for password-based authentication or authentication using public and private keys. The SSH suite of protocols is included with all modern UNIX and Linux distributions and with macOS client operating systems. For Windows-based computers, you need to download a freeware SSH client, such as PuTTY (putty.org).
- RDP (Remote Desktop Protocol) is a Microsoft proprietary protocol used by Windows Remote Desktop and Remote Assistance client-server utilities to connect to and control

a remote computer. RDP provides a remote desktop connection, which means that any information that would be displayed if you were sitting in front of the remote computer is instead displayed on your local computer's screen.

- Before the devices can be configured, they must already be booted up and they must already have configuration software installed. This is called in-band management, and it inherently limits troubleshooting capabilities. Out-of-band management, however, relies on a dedicated connection (either wired or wireless) between the network administrator's computer and each critical network device.
- A VPN (virtual private network) is a virtual connection that remotely accesses resources between a client and a network (client-to-site VPN), two networks (site-to-site VPN), or two hosts over the Internet or other types of networks (host-to-host VPN).
- To ensure a VPN can carry all types of data in a private manner over any kind of connection, special VPN protocols encapsulate higher-layer protocols in a process known as tunneling. Two common approaches to VPN tunneling either require all network traffic to traverse the VPN tunnel or only some of that traffic.
- While IPsec itself can be used as a tunneling protocol, it often provides encryption for another tunneling protocol. This pairing offers more flexibility and other features that IPsec can't offer alone.

Section 4-4: Troubleshooting Network Issues

- The netstat utility displays TCP/IP statistics and details about TCP/IP components and connections on a host. The netstat command can be used to obtain information about the port on which a TCP/IP service is running, which network connections are currently established for a client, how many messages have been handled by a network interface since it was activated, and how many data errors have occurred on a particular network interface.
- The Windows tracert utility uses ICMP echo requests to trace the path from one networked node to another, identifying all intermediate hops between the two nodes. Linux, UNIX, and macOS systems use UDP datagrams or, possibly, TCP SYN messages, for their traceroute utility, but the concept is still the same.
- The tcpdump utility is a free, command-line packet sniffer that runs on Linux and other UNIX operating systems. Like Wireshark, tcpdump captures traffic that crosses a computer's network interface. Because of its robust configuration options and straightforward, command-line interface, it's a popular tool among security professionals and hackers alike.
- Most of the time duplicate MAC addresses only cause intermittent connectivity issues for the computers involved in the duplication. It happens most often when managing multiple virtual devices on a large network, and in those cases, it's typically due to human error.
- When a router, switch, NIC, or other hardware goes down, a network technician's job is to identify the location of the hardware failure. Various tools help to narrow down the possibilities, including using tracert or traceroute to track down malfunctioning routers and other devices on larger networks, targeting a node on the other side of a suspected router problem rather than aiming for the router itself, and using ping to test for network connectivity.

Key Terms

For definitions of key terms, check the Glossary.

AH (authentication header)
ARP (Address Resolution Protocol)
ARP table
asymmetric encryption
authentication
CA (certificate authority)
CDP (Cisco Discovery Protocol)
checksum
CIA (confidentiality, integrity, and availability) triad
clientless VPN
client-to-site VPN
digital certificate
DoH (DNS over HTTPS)
DoT (DNS over TLS)
dynamic ARP table entry
encryption
ESP (Encapsulating Security Payload)
Ethernet II
full tunnel VPN
GRE (Generic Routing Encapsulation)
hop
in-band management
IPsec (Internet Protocol Security)
jumbo frame
key
key management
link-layer address
LLDP (Link Layer Discovery Protocol)
MTU (maximum transmission unit)
NDP (Neighbor Discovery Protocol)
netstat
OOBM (out-of-band management)
packet sniffer

PKI (public-key infrastructure)
private key encryption
probe
protocol analyzer
public key encryption
remote access
remote desktop connection
self-signed certificate
site-to-site VPN
split tunnel VPN
spoofing
static ARP table entry
symmetric encryption
tcpdump
terminal emulator
TFTP server
three-way handshake
traceroute
tracert
VPN (virtual private network)
VPN headend

Acronyms Checklist

The acronyms in Table 4-13 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 4-13 Chapter 4 acronyms

Acronym	Spelled out
AH	authentication header
CDP	Cisco Discovery Protocol
CIA	confidentiality, integrity, and availability
DoH	DNS over Hypertext Transfer Protocol Secure
DoT	DNS over Transport Layer Security
ESP	Encapsulating Security Payload
GRE	Generic Routing Encapsulation
IKE	Internet Key Exchange
IPsec	Internet Protocol Security
LLDP	Link Layer Discovery Protocol
MTU	maximum transmission unit
PKI	public-key infrastructure
SFTP	Secure FTP
TFTP	Trivial FTP
VPN	virtual private network

Review Questions

1. Which protocol's header would a layer 4 device read and process?
   a. IP
   b. TCP
   c. ARP
   d. HTTP
2. What field in a TCP segment is used to determine if an arriving data unit exactly matches the data unit sent by the source?
   a. Source port
   b. Acknowledgment number
   c. DiffServ
   d. Checksum
3. What is the Internet standard MTU?
   a. 65,535 bytes
   b. 1,522 bytes
   c. 1,500 bytes
   d. 9,198 bytes
4. Which two protocols manage neighbor discovery processes on IPv4 networks?
   a. ICMP and ARP
   b. IPv4 and IPv6
   c. TCP and UDP
   d. NDP and Ethernet

5. Which encryption protocol does GRE use to increase the security of its transmissions?
   a. SSL
   b. SFTP
   c. IPsec
   d. SSH
6. Which encryption benchmark ensures data is not modified after it's transmitted and before it's received?
   a. Confidentiality
   b. Integrity
   c. Availability
   d. Symmetric
7. Which remote file access protocol is an extension of SSH?
   a. SFTP
   b. TFTP
   c. FTPS
   d. HTTPS
8. Which of the following characteristics describes UDP?
   a. Establishes a session
   b. Emphasizes speed over accuracy
   c. Uses sequencing and checksums
   d. Provides flow control
9. Which secured tunneling protocol might be able to cross firewalls where IPsec is blocked?
   a. OpenVPN
   b. IKEv2
   c. L2TP
   d. GRE
10. Which command would display statistics about errors on a network interface?
   a. `tcpdump -D`
   b. `arp -d`
   c. `netstat -e`
   d. `traceroute -I`

Scenario-Based Questions

Question 4-1

You're getting a duplicate IP address error on your computer and need to figure out what other device on your network is using the IP address 192.168.1.56. What command will show you which MAC address is mapped to that IP address?

a. `telnet 192.168.1.56`
b. `tracert 192.168.1.56`
c. `arp -a`
d. `netstat -n`

Question 4-2

When surfing online, you get some strange data on an apparently secure website, and you realize you need to check the legitimacy of the site. What entity issues digital certificates for websites?

a. Registrar
b. CA
c. PKI
d. Active Directory

Question 4-3

You're working on a network performance issue after receiving several trouble tickets from users, and you need to check the configuration of a router in a nearby building. Which utility should you use to access the router securely over the network?

a. SSH
b. Telnet
c. SFTP
d. RDP

Question 4-4

You're having trouble authenticating to a web server on your local network and need to check the handshake conversation between your local device and the server. What `tcpdump` command will capture data on the enp0s3 interface and redirect output to a text file named mycapture.txt for further analysis?

a. `tcpdump -i enp0s3 mycapture.txt`
b. `tcpdump -i enp0s3 | mycapture.txt`
c. `tcpdump -i enp0s3 -w mycapture.txt`
d. `tcpdump -i enp0s3 -r mycapture.txt`

Hands-On Projects

Note 4-12 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 4-1: Redirect Command Output to a Text File

Estimated time: 15 minutes
Objective 5.5: Given a scenario, use the appropriate tool or protocol to solve networking issues.
Resources:
- Internet access

Description:
Sometimes when you're using a command such as `tcpdump`, the sheer volume of output can be daunting to work with. There's no way to search through the output for specific information, and you can only expand the PowerShell or Command Prompt window so far. One solution to this problem is to redirect the command output to a text file where you can search or sort the text, copy and paste text, and save the output for future reference. To accomplish this feat, you'll need to add a redirection operator to the command whose output you want to export to a text file. Complete the following steps:

1. First, enter this simple command in PowerShell or Command Prompt:

   `ipconfig > ipconfigtest.txt`

   This runs the `ipconfig` command and redirects the output to a text file named ipconfigtest.txt. By default, the file is saved to the current default folder, for example, C:\Users\Your Name. Use File Explorer to find the file. **Take a screenshot** showing the file and its file path; submit this visual with your responses to this project's questions.

2. To specify the location of the file when you create it, add the path for the file in the command line. For example, to save the file to the desktop, use the following command (substitute the correct file path to your desktop). If you have a space in your username, be sure to surround it with quotation marks, such as `\"Your Name"\`. What command and file path did you enter?

   `ipconfig > C:\Users\`*`Username`*`\Desktop\ipconfigtest.txt`

Note 4-13 | If you're not sure what the file path is to your Desktop, you can find it in File Explorer. In the navigation pane on the left, right-click the **Desktop** link and click **Properties**. The file path is shown on the Location tab. Note that your desktop might be showing your OneDrive Desktop, which would be located at the following path:

C:\Users*Username*\OneDrive\Desktop

In this case, you might not be able to send command output to your OneDrive desktop. Save your files for this project to a folder on your hard drive instead.

3. Open the file and view the contents. How many active network connections are listed (that is, how many network connections list an IPv4 and/or an IPv6 address)?
4. If you already have a file on the desktop by that name, the file will be overwritten with the new data. What if you would rather append data to an existing file? In this case, use the `>>` operator. Close the ipconfigtest.txt file. Then enter this command in the terminal (substitute the correct file path to your desktop):

   ```
   ipconfig >> C:\Users\Username\Desktop\ipconfigtest.txt
   ```

Note 4-14 | When reusing an earlier command or portions of an earlier command, you can press the up arrow on your keyboard to recall earlier commands. Then use the side arrows to place your cursor to make edits.

5. Open the file. How many active network connections are listed now? Notice the new output appears at the end of the existing file, and all the data is preserved within this single file. This option is useful when collecting data from repeated tests or from multiple computers, where you want all the data to converge into a single file for future analysis.
6. Where do command parameters fit when redirecting output? Let's try the `netstat` command to show the IP address and port of each TCP and UDP connection on the computer. Make sure all browsers on your computer are closed. In the following command, substitute the correct file path to your desktop to output the data to a new file. Notice that any parameters you want to use should be inserted after the command itself and before the redirection operator.

   ```
   netstat -an > C:\Users\Username\Desktop\connections.txt
   ```

7. Open the file you just created. How many TCP ports are in the ESTABLISHED state?
8. Open your browser and visit two or three websites in different tabs. With your browser still open, run the same `netstat` command again, but this time, append the new data to your existing file. What command did you run? How many TCP ports are in the ESTABLISHED state now?
9. You can include a space in the filename by putting quotation marks around the portion of the path containing the space or around the entire filename and location. Enter the following command:

   ```
   ping 8.8.8.8 > "C:\Users\Username\Desktop\find google.txt"
   ```

10. **Take a screenshot** of your File Explorer window showing the files you created in this project from Step 2 through Step 9; submit this visual with your responses to this project's questions.

Project 4-2: Repair a Duplicate IP Address

Estimated time: 15 minutes

Objective 5.3: Given a scenario, troubleshoot common issues with network services.

Resources:

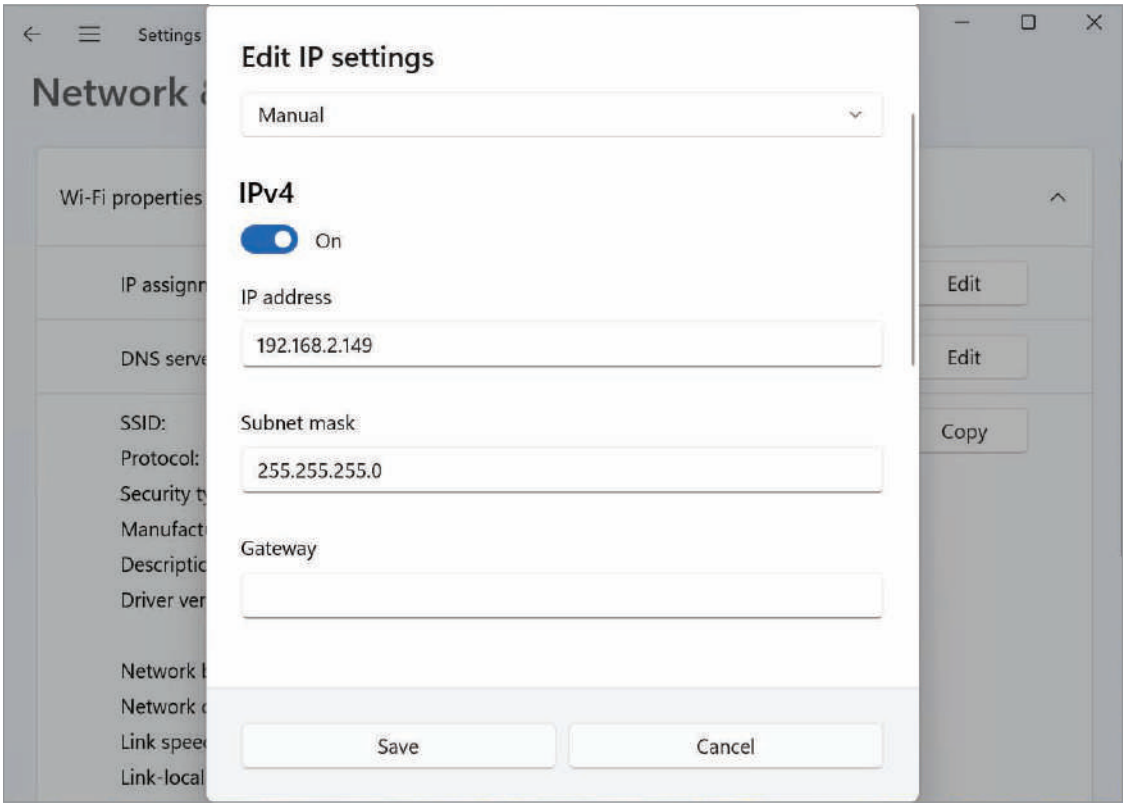
- Windows computer with administrative access

Description:

ARP can be a valuable troubleshooting tool for discovering the identity of a machine whose IP address you know, or for identifying two machines assigned the same IP address. Let's find out what happens when two devices on the network are assigned the same IP address. First you change the IP address of a local Windows machine to match an IP address of another device–in other words, you "break" the computer's connection to the network. Then you see how the `arp` command helps you diagnose the problem. Complete the following steps:

1. Open a PowerShell or Command Prompt window and enter the command `arp -a`. Your device's IP address is listed as the Interface address at the top of the list. Write down this IP address and the address of another device on the network.
2. In the Settings app, click **Network & internet**, click the active network connection, and then click **Hardware properties**. Next to IP assignment, click **Edit**.
3. Make sure the IP settings are set to **Manual**. Set the IP address to match the other device's IP address that you wrote down in Step 1. For a home network, most likely the subnet mask is 255.255.255.0 (Figure 4-27). Click **Save** and then close the Settings app.
4. Back at the CLI, enter `ipconfig /all`.

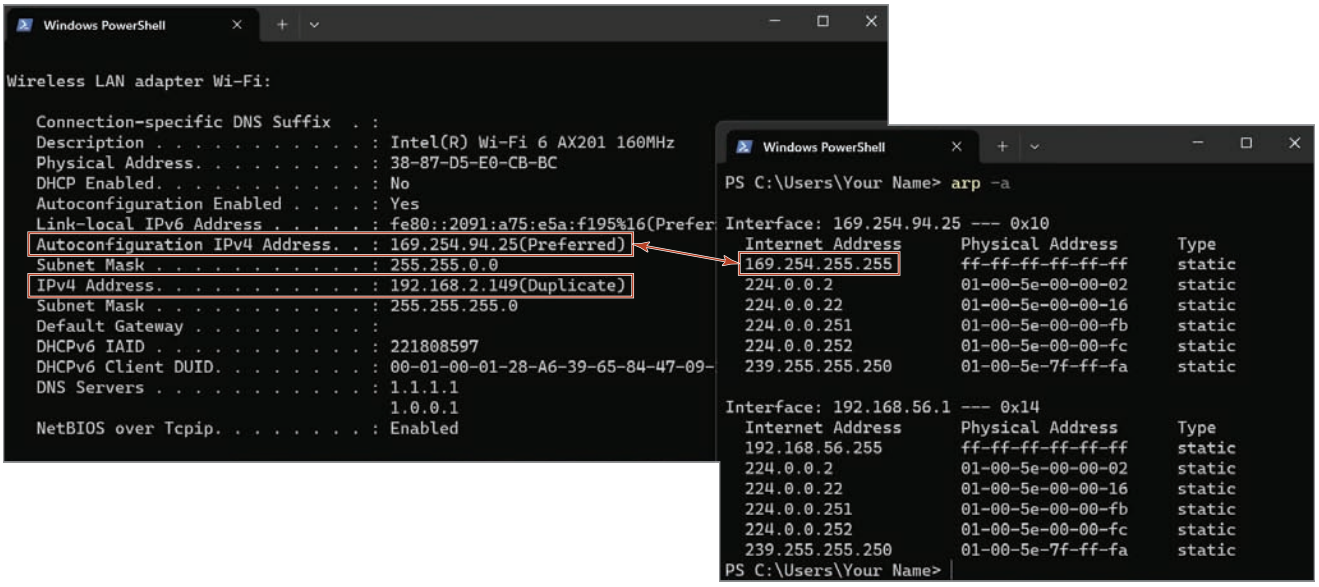
Figure 4-27 Set the IP address and subnet mask



Source: Microsoft Corporation

5. Find the appropriate network connection and identify your computer's current IPv4 address. Respond to the following questions:
   a. Has your computer identified the duplicate IP address problem yet? How do you know?
   b. Your computer might also have autoconfigured another IP address. If so, what address did your computer resort to?
   c. **Take a screenshot** of your TCP/IP configuration information; submit this visual with your responses to this project's questions.
6. In the left window in Figure 4-28, a warning identifies that the IP address is a duplicate. The system also lists a preferred IPv4 address (in this case, 169.254.94.25), which is an APIPA address. How can you tell this is an APIPA address?
7. Enter the command `arp -a`. In the right window in Figure 4-28, the computer only lists its own APIPA address and no longer includes other devices on the network. What are two ways to solve this problem of a duplicate IP address?

Figure 4-28 The computer automatically configured an APIPA address



Source: Microsoft Corporation

8. In Settings, return to the active connection's hardware properties page, click to edit the IP assignment, and select **Automatic (DHCP)** from the list. Then click **Save.** Close all active windows except your terminal.
9. Run the `ipconfig` command or the `arp -a` command to confirm that a unique IP address has been assigned to your local device's active network interface. What is the new IP address?
10. Close the terminal window.

Capstone Projects

Note 4-15 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 4-1: Set Up and Use a TFTP Server in Packet Tracer

Estimated time: 30 minutes

Objective 3.1: Explain the purpose of organizational processes and procedures.

Resources:

- Computer with Cisco Packet Tracer installed

Description:

In Capstone Project 2-2, you installed Packet Tracer, and you used it again in Capstone Project 3-2. Earlier in this chapter, you learned about TFTP servers that can be used to back up and configure network devices. In this Capstone Project, you configure a router, back up the router configuration on a TFTP server, and then create a replacement router from the backup file. Note that Cisco devices keep active configuration settings in a file called running-config. There are actually two files: The startup-config file provides initial settings when the device boots, and the running-config file maintains those settings in the device's memory while it is powered on. When you make changes to the running-config file, this edits do not change the startup-config file, which means your settings are lost on the device's next power cycle. You can copy the running-config file contents to the startup-config file for later access after a power cycle. In this project, you'll focus on working with the running-config file. Complete the following steps:

1. Open Cisco Packet Tracer and, if necessary, sign in with your Skills for All account.
2. In the Devices pane, click the **Network Devices** category and then click **Routers**. Add a **PT-Router** to your workspace. Below the router's icon in the workspace, change the router's display name to **RtrGiraffe**. Add a note to indicate the router's IP address: **192.168.2.1/24**
3. Click the router to open its configuration window. Click the **Config** tab. Change the hostname to **RtrGiraffe**. After you make this change, notice the commands that scroll in the Equivalent IOS Commands pane at the bottom of the window.
4. Under INTERFACE in the left pane, click **FastEthernet0/0**. Set the router's IPv4 address to **192.168.2.1**. The subnet mask field should automatically populate with 255.255.255.0 when you tab to it. Check the **On** box to activate the port, and then close the router's configuration window.
5. In the Devices pane, click the **End Devices** category. Add a **Server** to your workspace. Below the server's icon in the workspace, change the server's display name to **TFTPserver**. Add a note to indicate the server's IP address: **192.168.2.100/24**
6. Click the server to open its configuration window. Click the **Config** tab and click **FastEthernet0** in the left pane. Set the server's IPv4 address to **192.168.2.100**. The subnet mask field should automatically populate with 255.255.255.0 when you tab to it. Close the server's configuration window.

7. In the Devices pane, click the **Connections** category. Because you're connecting a server directly to a router, you need a crossover cable, which you'll learn more about later in this course. Click the straight, dashed line for the **Copper Cross-Over** cable and connect the **FastEthernet0/0** interface on the router with the **FastEthernet0** interface on the server.
8. In the server's configuration window, click the **Desktop** tab and click **Command Prompt**. Ping the router to confirm the connection works. What command did you use? Close the server's configuration window.

The server's TFTP service is on by default, so you're now ready to back up the router's running-config file to the server. You'll need to perform this task from the router's CLI. Complete the following steps:

9. In the router's configuration window, click the **CLI** tab. Click inside the IOS Command Line Interface pane and press **Enter**. The prompt should show `RtrGiraffe(config-if)#`. This prompt gives you some helpful information:
   a. The router's hostname is RtrGiraffe.
   b. The router's CLI is currently in the interface configuration mode, which is what you used to configure the FastEthernet0/0 interface. To exit this mode, enter `exit`.
10. Your router is now in global configuration mode, indicated by the `(config)#` portion of the prompt. You need to step down one more mode level, so enter `exit` again and then press **Enter** to return to the prompt.
11. Your router is now in privileged EXEC mode, which is what you need to copy the running-config file to the TFTP server. Enter the command `copy running-config tftp`, which instructs the router to copy its running-config file to the TFTP server. You'll need to provide a little more information:
    a. Enter the server's IP address: **192.168.2.100**
    b. Enter the name of the file to be saved on the server: **Giraffe**
12. Close the router's configuration window.

Now you're ready to check the logs on the TFTP server to confirm the file was copied successfully. Complete the following steps:

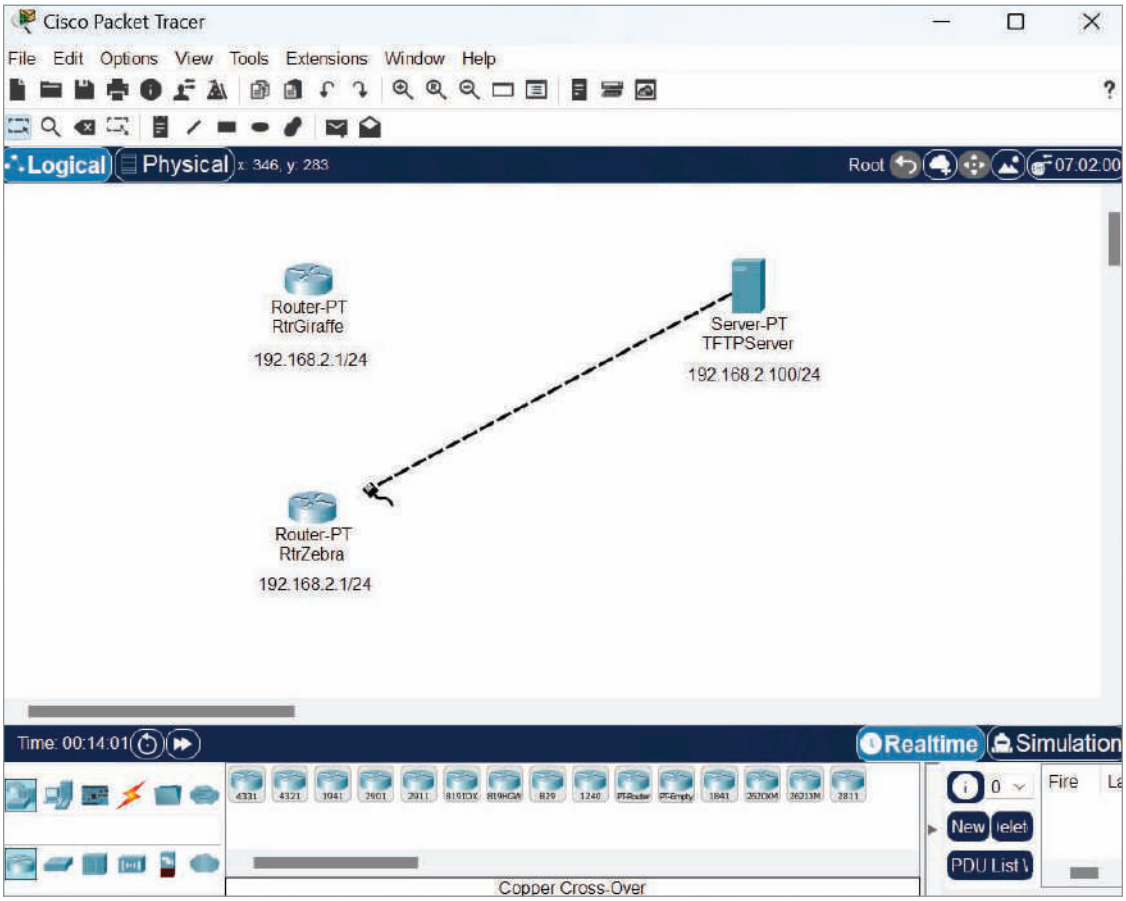
13. In the server's configuration window, click the **Services** tab and then click **TFTP** in the left pane. **Take a screenshot** of the logs, which should show the Giraffe entry at the top. Submit this visual with your responses to this project's questions. Close the server's configuration window.

Suppose your existing router fails, and you need to replace it with a new router. In the real world, you likely would have made many more configuration changes than simply the router's hostname and IP address. It would be time-consuming and risky to attempt to copy this configuration manually to a new router. But you are prepared and have a backup configuration file! You're now ready to create a new router and copy the configuration file to this router. Complete the following steps:

14. From the Devices pane, add a new **PT-Router** to your workspace. Give this router the display name *and* the hostname **RtrZebra**, and set the FastEthernet0/0 interface to the same IP address as the first router: **192.168.2.1**. Be sure to make these changes to the documentation in your workspace *and* to the router's configuration settings. The workspace documentation is for your benefit, but the router only knows about changes made within its configuration window. Also remember to activate the interface.

15. Grab RtrGiraffe's end of the cable and drag it to RtrZebra, connecting the cable to the **FastEthernet0/0** interface (Figure 4-29). From the server, ping the new router to confirm the connection works. What command did you use?

Figure 4-29 Move the cable from the old router to the new router



Source: Cisco Systems, Inc.

16. In RtrZebra's configuration window, click the **CLI** tab. Click inside the IOS Command Line Interface pane and press **Enter**. What is this router's hostname, as indicated by the prompt?
17. Enter the `exit` command twice and press **Enter** again to get to privileged EXEC mode. Confirm the prompt says `RtrZebra#` before continuing.
18. To copy the old router's running-config file from the TFTP server to the new router, enter the command `copy tftp running-config`. Provide the TFTP server's IP address and the filename to be copied (**Giraffe**). Press **Enter** twice more. What is the new router's hostname now, as indicated by the prompt? Why do you think this is?
19. At this point, if you were to power cycle the new router, it would lose all these settings. To save the running-config data to the startup-config file so the changes persist, enter the command `copy running-config startup-config` and then press **Enter** when prompted.
20. You do not need to save this Packet Tracer network for future projects. Before closing the network, take some notes in your Wikidot website about your work in this project, commands that you learned, and new insights you now have about how Packet Tracer works.

Note to Instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site.

Capstone Project 4-2: Use SSH in Ubuntu

Estimated time: 30 minutes (+5 minutes for group work, if assigned)
Objective 3.5: Compare and contrast network access and management methods.
Group work: This project includes enhancements when assigned as a group project.
Resources:
- Access to the same computer used to complete Capstone Project 1-1.
- Internet access

Description:
In this project, you will learn to use SSH in Ubuntu. Using the Ubuntu VMs you created in Capstone Projects 2-1 and 3-1, follow these steps to create an SSH connection.

1. When using VirtualBox, you first need to check the Network settings for both VMs. In the VirtualBox window, select each VM, click **Settings**, and click **Network**. If necessary, change the *Attached to* setting to **Bridged Adapter**. Click **OK**.

On the Ubuntu Server VM, do the following:

2. Start the VM and log on. Refer to your Bitwarden vault if you don't remember your logon information.
3. SSH is included in Ubuntu Server but is not installed. First update the apt database with the command `sudo apt update`, and then enter your password to complete the update. To install and start SSH, enter the command `sudo apt install ssh` and then give permission to install the software.
4. Enter the command `ip address show` and write down the IP address of the Ubuntu Server VM. Leave this VM running.

On the Ubuntu Desktop VM, do the following:

5. Start the VM and log on. Refer to your Bitwarden vault if you don't remember your logon information.
6. Open Terminal and enter the command `ip address show`. Note the IP address of the Ubuntu Desktop VM. What is the command prompt? Given this information, which VM are you interacting with?
7. Enter the `hostname` command. Which VM's host name is listed?
8. Enter the `ssh` command with the IP address of the Ubuntu Server VM. For example, if the server IP address is 192.168.1.147, enter this command:

   ```
   ssh 192.168.1.147
   ```

   If your username on the Ubuntu Server machine is not the same as your username on the Ubuntu Desktop machine, you'll need to add a bit more information to this command to remote into the server. Try this command instead:

   `ssh` *`server_username@server_ipaddress`*

   For example, if the server IP address is 192.168.1.147 and the server username is yourname, you would enter this command:

   ```
   ssh yourname@192.168.1.147
   ```

9. Enter `yes` to confirm you want to establish the connection, and then enter your password to log on to the server using SSH. You now have an SSH session established between the Ubuntu Desktop VM and the Ubuntu Server VM. What is the command prompt? Given this information, which VM are you interacting with?

10. Enter the `hostname` command. Which VM's host name is listed?
11. Enter the `dir` command. What directory is listed? Recall that you created this directory in Capstone Project 3-1.
12. Enter the `ip address show` command. Which IP address is displayed in the command output: the Ubuntu Desktop VM's address or the Ubuntu Server VM's address?
13. **Take a screenshot** of your Ubuntu Desktop VM's Terminal window showing your commands run on the SSH session with the Ubuntu Server VM; submit this visual with your responses to this project's questions.
14. When you're finished using the SSH session, enter the `exit` command to close the session.
15. **For group assignments:** Establish an SSH session with another group member's Ubuntu VM, either their Desktop VM or their Server VM. The VM's owner will need to enter their credentials to authenticate the connection. What command did you use to establish the connection?
16. To shut down the VMs, enter the `sudo poweroff` command for each VM and, if needed, your password.
17. Add some notes to your Wikidot website about the SSH installation on the Ubuntu Server VM.

Solutions to Self-Check Questions

Section 4-1: TCP/IP Core Protocols

1. Which protocol's header includes the source MAC address?
   Answer: a. Ethernet
   Explanation: **Ethernet** encapsulates data with a header and trailer, creating a frame. This frame includes the MAC address of the sender. A header for TCP and UDP at the transport layer includes a port to identify the communicating application on the source host. IP identifies the IP address of the source host.

2. Which of these protocols requires the transport layer header to include a checksum field?
   Answer: c. IPv6
   Explanation: **IPv6** requires that the transport layer header (TCP or UDP) include a checksum field because IPv6 does not include its own checksum field. IPv4 and ICMP both include a checksum field as well, either for the header or for the entire packet.

3. An ARP table maps MAC addresses to what information?
   Answer: d. IPv4 addresses
   Explanation: ARP (Address Resolution Protocol) works in conjunction with IPv4 to discover the MAC address of a node on the local network and to maintain a database that maps local **IPv4 addresses** to MAC addresses. A MAC address table maps MAC addresses to physical interfaces on a switch. IPv6 relies on ICMPv6 to discover other nodes on a network. Gateways such as routers track TCP and UDP ports for address translation.

Section 4-2: Encryption Protocols

4. Which component of the CIA triad can be ensured when data is encrypted by a private key and decrypted by the matching public key?
   Answer: d. Integrity
   Explanation: Public key encryption can ensure data **integrity**, as the sender's public key will only work to decrypt data encrypted with the sender's private key if the data has not been tampered with. Encrypting data with a public key and decrypting with the matching private key ensures confidentiality, as only the intended recipient (the owner of the keys) can use the private key to decrypt data encrypted with their corresponding public key. Availability refers to the data being available to the intended recipient when needed, meaning the sender is accountable for successful delivery of the data.

5. Which IPsec encryption type encrypts the IP header?
   Answer: b. ESP
   Explanation: Either AH (authentication header) encryption or ESP (Encapsulating Security Payload) encryption may be used in IPsec. Both types of encryption provide authentication of the IP packet's data payload through public key techniques. In addition, **ESP** encrypts the entire IP packet, including the IP header. IKE (Internet Key Exchange) negotiates the exchange of keys, including authentication of the keys. ISAKMP (Internet Security Association and Key Management Protocol) works within the IKE process to establish policies for managing the keys.

Section 4-3: Remote Access Protocols

6. Which remote file access protocol uses port 22?
   Answer: d. SFTP
   Explanation: Unlike FTP (File Transfer Protocol) or FTPS (FTP Secure), which use a control channel on port 21 and one or two data channels on port 20, **SFTP (Secure FTP)** uses only a single connection—both inbound and outbound communications are usually configured to cross SSH's port 22. TFTP (Trivial FTP) listens at port 69.

7. You need to remote into a Linux server in another building on your network. Which of the following protocols should you use?
   Answer: b. SSH
   Explanation: The **SSH (Secure Shell)** suite of protocols is included with all modern UNIX and Linux distributions and with macOS client operating systems. RDP (Remote Desktop Protocol) is a Microsoft proprietary protocol used by Windows. SFTP (Secure FTP) is a file-transfer version of SSH that includes encryption and authentication. A VPN (virtual private network) is a network connection encrypted from end to end that creates a private connection to a remote network, not a server in another portion of the same network.

8. You're working from home and need to access a file server at the office while working in an application from your work desktop. At the same time, you often stream music in your browser. Which VPN type will be most efficient while still meeting your needs?
   Answer: d. Split tunnel VPN
   Explanation: A **split tunnel VPN** will only capture traffic destined for the corporate network, such as communication with the file server and the work desktop, and the client can use local network resources to stream music from the Internet, which will reduce latency and will use fewer corporate network resources. A full tunnel VPN captures all network traffic, whether destined for the Internet or for the corporate network. A host-to-host VPN creates a VPN tunnel directly between two devices and would not provide access to other resources, such as music streaming. A site-to-site VPN connects multiple sites across a WAN and requires more expensive equipment than a client-to-site VPN for the split tunnel configuration.

Section 4-4: Troubleshooting Network Issues

9. You need to determine which device on your network is sending excessive messages to your Ubuntu Server. Which utility will give you this information?
   Answer: b. tcpdump
   Explanation: The **tcpdump** utility captures traffic that crosses a computer's network interface, which can then be analyzed to determine which devices are transmitting traffic across each interface. The traceroute utility traces the path from one networked node to another, identifying all intermediate hops between the two nodes. The netstat utility displays TCP/IP statistics and details about TCP/IP components and connections on a host. The `arp` command provides a way of obtaining information from and manipulating a device's ARP table.

10. What protocol must be allowed through a firewall for tracert to work correctly?
    Answer: c. ICMP
    Explanation: The Windows tracert utility sends an **ICMP (Internet Control Message Protocol)** echo request to the destination node and listens for an ICMP echo reply from that node. SSH (Secure Shell) is a collection of protocols that performs both authentication and encryption. NDP (Neighbor Discovery Protocol) information carried in ICMPv6 messages automatically detects neighboring devices and automatically adjusts when neighboring nodes fail or are removed from the network. TLS (Transport Layer Security) is a method of encrypting TCP/IP transmissions en route between the client and server using public key encryption technology.